Centre Number	Candidate Number	Candidate Name

NAMIBIA SENIOR SECONDARY CERTIFICATE

CHEMISTRY ORDINARY LEVEL 6117/3

PAPER 3 Alternative to Practical — 1 hour 15 minutes

Marks 40 — **2023**

Additional materials: Non-programmable calculator
Ruler

INSTRUCTIONS AND INFORMATION TO CANDIDATES

- Candidates answer on the Question Paper in the spaces provided.
- Write your Centre Number, Candidate Number and Name in the spaces at the top of this page.
- Write in dark blue or black pen.
- You may use a soft pencil for any diagrams, graphs or rough working.
- Do not use correction fluid.
- Do not write in the margin *For Examiner's Use*.

- Answer **all** questions.

- The number of marks is given in brackets [ ] at the end of each question or part question.
- You may use a non-programmable calculator.

For Examiner's Use	
1	
2	
3	
4	
TOTAL	

Marker	
Checker	

This document consists of **10** printed pages and **2** blank pages.

Republic of Namibia

MINISTRY OF EDUCATION, ARTS AND CULTURE

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1 Fig. 1.1 shows a container of a cleaning agent, that contains ammonia and sand.

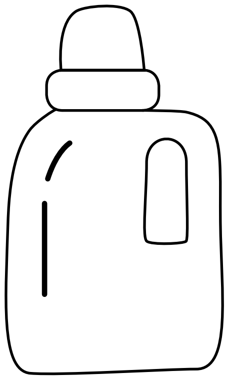


Fig. 1.1

(a) Describe a test to show that the sample of the cleaning agent contains ammonia.

test ....

result .... [2]

(b) Fig. 1.2 shows the apparatus used to obtain pure water from the cleaning agent mixture.

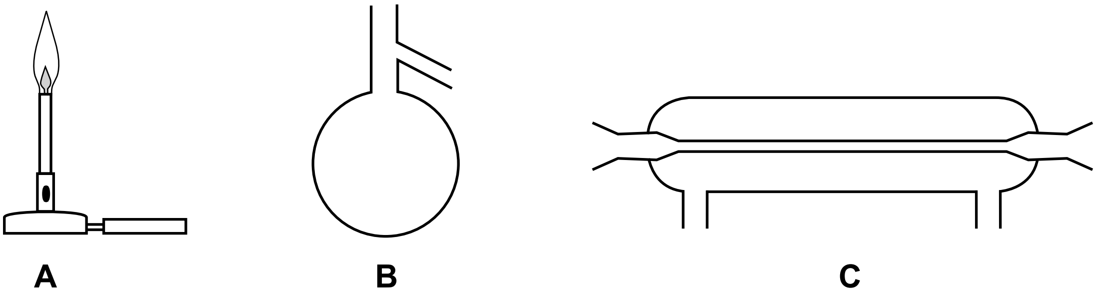


Fig. 1.2

Name the apparatus **A**, **B** and **C** shown in Fig. 1.2.

A ....

B ....

C .... [3]

(c) Explain how to obtain pure sand from the cleaning agent mixture.

....

....

....

....

....

.... [3]

[8]

2 Calcium carbonate reacts with dilute hydrochloric acid releasing carbon dioxide gas. Fig. 2.1 shows the set-up of the experiment.

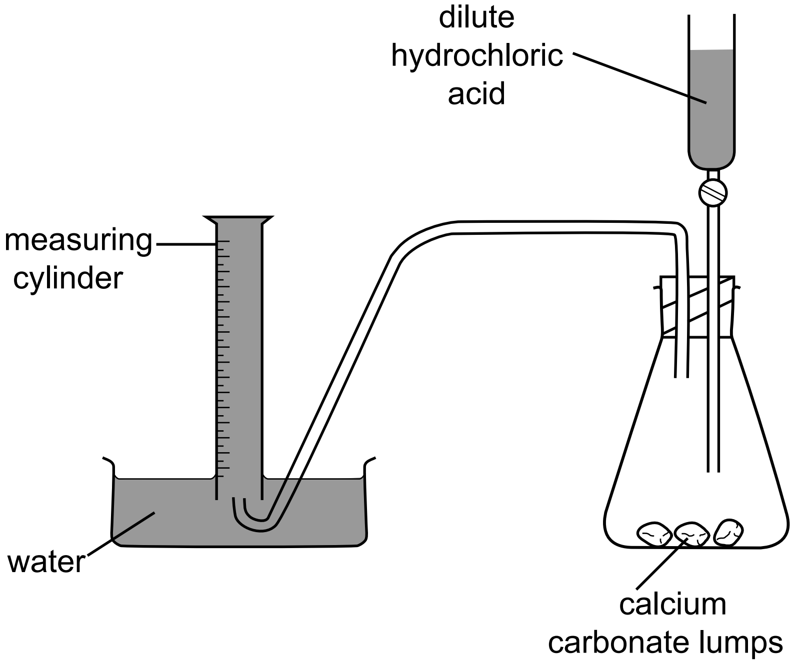


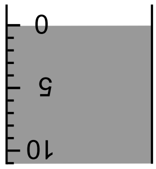
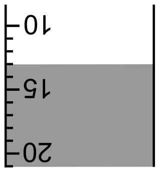
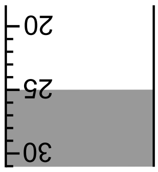
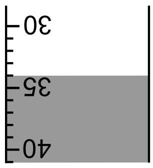
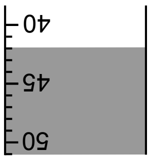
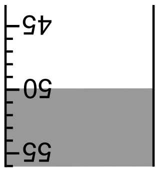
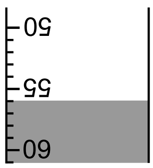
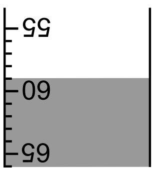
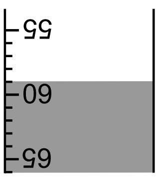
Fig. 2.1

Large lumps of calcium carbonate were used.

The volume of carbon dioxide produced was measured every 30 seconds for 240 seconds.

(a) Use the measuring cylinder diagrams to complete Table 2.1.

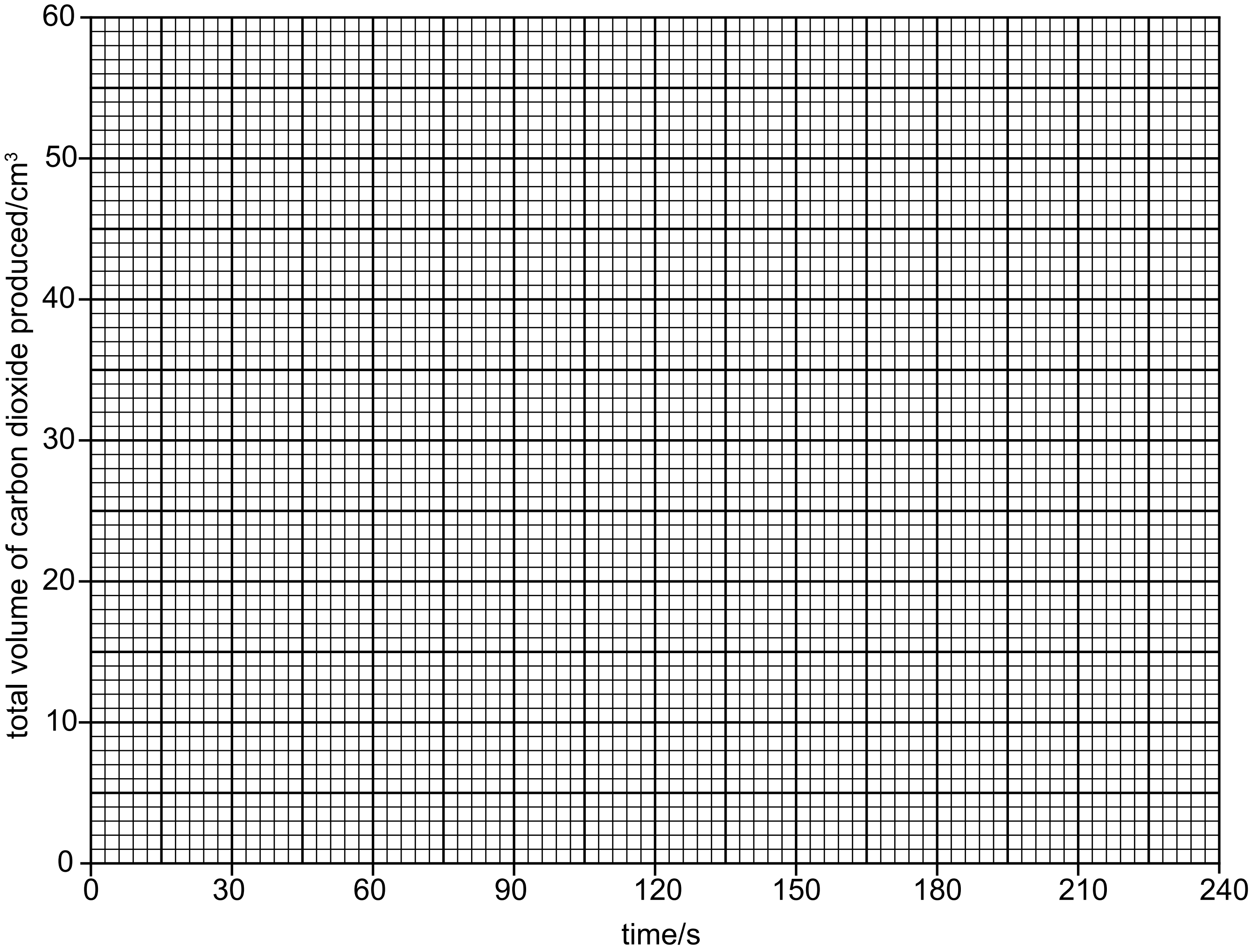
Table 2.1

time / s	measuring cylinder diagram	total volume of carbon dioxide produced / cm^3
0		
30		
60		
90		
120		
150		
180		
210		
240		

[3]

(b) Use the data in Table 2.1 to plot a graph.

Draw a line of best fit.



[3]

(c) The average rate of this reaction can be calculated by using the equation

$$\text{average rate} = \frac{(\text{total volume of carbon dioxide produced / cm}^3)}{(\text{time / s})}.$$

Calculate the average rate of this reaction for the first 100 seconds.

Include units.

average rate ....

unit .... [3]

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(d) State a reason why, eventually, no more gas will be produced.

.... [1]

(e) Suggest and explain the effect on the rate of reaction of using the same mass of calcium carbonate powder instead of large lumps of calcium carbonate.

....

....

....

.... [2]

(f) One learner suggests that using a gas syringe to collect the gas instead of a measuring cylinder will improve the accuracy of the experiment.

State whether you agree or disagree with the statement and justify your answer.

statement ....

justification ....

.... [2]

[14]

3 Two solids, **A** and **B**, were analysed. Solid **B** was lithium chloride. Tests were carried out on each solid. Some of the observations of solid **A** are shown.

tests on solid A	observations
Appearance of solid **A**.	blue crystals
Test 1 Part of solid **A** was placed in a boiling tube and heated gently, until the blue colour has disappeared.	the solid turned white liquid droplets formed in the boiling tube
Test 2 The liquid droplets which formed in the boiling tube were tested with universal indicator paper.	the universal indicator paper turned green
The remaining solid **A** was placed in a test-tube and about 10 cm^3 of distilled water was added to the test-tube. The mixture was shaken to dissolve solid **A** to form solution **A**. Solution **A** was divided into two test-tubes. **Test 3** Aqueous sodium hydroxide was added to the first portion of solution **A**.	light blue precipitate, insoluble in excess
Test 4 Dilute nitric acid was added to the second portion of solution **A**, then aqueous barium nitrate was also added.	white precipitate
Test 5 A flame test was carried out on solid **A**.	blue-green colour

(a) **Test 1** states that the solid should be heated gently, then strongly.

In terms of safety, explain why it is necessary to heat gently at first.

....

.... [1]

(b) Identify the liquid being tested in **test 2**.

.... [1]

(c) Identify solid **A**.

.... [2]

(d) The container of nitric acid is labelled corrosive.

State **one** safety precaution that should be taken when using nitric acid.

....

.... [1]

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Tests on solid B

Complete the expected observations.

(e) Describe the appearance of solid **B**.

.... [1]

(f) Distilled water was added to solid **B** in a test-tube and shaken to dissolve solid **B** to form solution **B**.

(i) To the first portion of solution **B**, an excess of aqueous sodium hydroxide was added.

observations.. [1]

(ii) To the second portion of solution **B**, dilute nitric acid and aqueous silver nitrate were added.

observations.. [2]

(g) A flame test was carried out on solid **B**.

observations .... [1]

[10]

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4 Sodium thiosulfate reacts with dilute hydrochloric acid. Sulfur forms during this reaction and the mixture turns cloudy.

$$Na_2S_2O_3(aq) + 2HCl(aq) \rightarrow 2NaCl(aq) + 2H_2O(l) + SO_2(g) + S(s)$$

Plan an experiment to investigate whether the speed of reaction between sodium thiosulfate and dilute hydrochloric acid depends on the concentration of sodium thiosulfate.

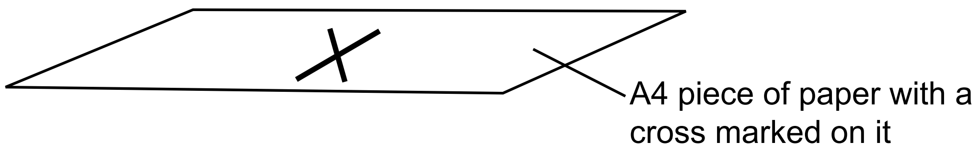
The following materials are provided.

- 50 cm^3 sodium thiosulfate of 0.25 mol/dm^3 concentration
- 50 cm^3 sodium thiosulfate of 1.00 mol/dm^3 concentration
- dilute hydrochloric acid
- 20 cm^3 measuring cylinder
- 2 × 250 cm^3 conical flasks
- 1 × stop-watch
- 1 × A4 piece of paper with a cross marked on it

You should:

Describe and explain how you would carry out the investigation, including the apparatus you would use.

- State the measurements you would take.
- State the variables you would control.
- Explain how you will use your results to reach a conclusion.



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[8]

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